Homework 1: Financial Data Analysis

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Due on February 13, 2026 (11:59pm)

Submit an electronic version of your answer (report and codes files) to Blackboard (no late submission will be accepted).

1. Import the data of Bitcoin price (adjusted close price) from the attached file. Calculate the daily log returns, and report the summary statistics of the log returns (mean, standard deviation, skewness, kurtosis). Apply the Shapiro-Wilk test and qq-plot to check the normality of the log returns. Comment on the distributional properties of the log Bitcoin returns.

2. Find the best-fit distribution for the log Bitcoin returns, report the parameters of the distribution estimated by the maximum likelihood or moment matching methods, and validate the goodness of fit by the Kolmogorov-Smirnov test and qq-plot. Try at least three candidate distributions.

3. Suppose you are holding a Bitcoin. Estimate the VaR and CVaR of the Bitcoin returns at various significance levels using the historical simulation method and the parametric method (i.e., first estimate a parametric distribution and then calculate the VaR and CVaR). Comment on the differences between the VaR and CVaR estimated by different methods.

4. Import the data of S&P 500 index options from the attached file. Follow the filtering process in the lecture notes and use the remained OTM options to construct the implied volatility curve. (The data is on 2025-02-13 and all these options will expire on 2025-03-13.)

5. Calculate the VIX index from these S&P 500 index options and compare it to the VIX index at 15.10 reported by the CBOE on 2025-02-13.

6. Extract mean, standard deviation, skewness, and kurtosis of the log returns of the S&P 500 index $\ln(S_T/S_t)$ (with T being the maturity date of the option and t being the current date (2025-02-13)) under the risk-neutral probability measure. Compare the distributional properties of the log returns of the S&P 500 index under the real-world probability measure and the risk-neutral probability measure (also use monthly return).

7. (Optional) Find the best-fit distribution for the log S&P 500 index returns $\ln(S_T/S_t)$ under the risk-neutral probability measure that minimizes the difference between the theoretical and market option prices.